BEEE301L– POWER ELECTRONICS

Course Project Report

BOOST PFC CONVERTER: DESIGN AND SIMULATION WITH ANN-BASED CAPACITOR HEALTH MONITORING

Boost PFC front-end simulated and demonstrated with a 10 W hardware prototype

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ABSTRACT

On-board chargers (OBCs) of electric vehicles draw power from a single-phase grid that is often unstable, and their failures are usually discovered only after a protective shutdown, by which time the sealed module must be replaced as a whole. This project addresses both the power-quality and reliability aspects of this problem. The front end of an electric vehicle charger is studied using a single-phase boost power factor correction (PFC) converter operating in continuous conduction mode. The converter boosts the rectified input voltage to a regulated DC link while improving the input power factor by shaping the line current to follow the line voltage.

The converter is modelled in MATLAB/Simulink to study its electrical behaviour and performance under different operating conditions. The simulation focuses on the input and output waveforms, DC-link voltage, inductor current, output-voltage ripple, and the response of the converter to changes in operating conditions. These results provide a basis for evaluating the converter's power-quality performance and understanding the effect of component variations on its operation.

A low-voltage hardware prototype keeps the same topology and control at 10 W, stepping 12 VAC up to 24 V DC at about 65 kHz with a Texas Instruments UCC28180 controller on a breadboard.

In addition to the converter study, an artificial neural network (ANN)-based method is implemented in MATLAB to estimate DC-link capacitor health from simulated output-voltage waveforms. A dataset is generated by varying the capacitance from 120 µF to 60 µF across 41 simulation cases. Five features are extracted from the output-voltage waveform: ripple, mean, standard deviation, maximum value, and minimum value. A feedforward neural network with eight hidden neurons is trained to estimate the capacitance. The estimated capacitance is then compared with a 96 µF threshold, corresponding to 80% of the nominal 120 µF capacitance, to flag potential capacitor degradation. Simulink tests demonstrate the method's response at selected capacitance values, including 120 µF, 100 µF, and 80 µF. This approach demonstrates how simulation data and neural-network-based estimation can support capacitor condition monitoring in a boost PFC converter.

Keywords: boost PFC converter, continuous conduction mode, EV on-board charger, UCC28180, power factor, total harmonic distortion, MATLAB/Simulink, artificial neural network, capacitor degradation detection, capacitance estimation, condition monitoring.

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CHAPTER 1: INTRODUCTION

1.1 Background

Electric vehicles sold in India are normally charged at home or at the workplace through an on-board charger connected to a single-phase socket. The charger has two jobs: it must convert alternating current into the direct current needed by the traction battery, and it must do so without polluting the supply. Any equipment that draws current only near the peaks of the voltage wave behaves like a harmonic source; it raises the RMS current for a given amount of real power, heats distribution transformers, and distorts the voltage seen by neighbouring loads. Standards such as IEC 61000-3-2 therefore cap the harmonic currents that equipment rated up to 16 A per phase may inject [15], and conductive charging equipment is additionally governed by IEC 61851-1 [16].

The usual way of meeting these limits is to place an active PFC stage directly after the diode bridge, followed by an isolated DC–DC converter that applies the constant-current/constant-voltage (CC/CV) profile to the battery [2], [18]. At the 3.3 kW level, which corresponds to a 230 V, roughly 16 A household outlet, the single-switch boost converter operating in continuous conduction mode (CCM) remains the reference PFC topology because it is simple, well documented and supported by dedicated controller ICs [1], [4].

Power quality is only part of the picture. Field surveys of converter reliability rank electrolytic capacitors, semiconductor devices and their solder and thermal interfaces among the most failure-prone parts of a power converter [10], and capacitor wear-out is accelerated by heat [9]. Indian operating conditions, with high ambient temperatures and frequent voltage swings, make this thermo-mechanical stress worse. Present chargers react to faults by tripping, which protects the vehicle but gives the owner no warning. The ideas of a digital twin, a live software model that mirrors a physical asset [11], and of data-driven condition monitoring [19], [20] suggest a different approach: watch the charger continuously, compare its behaviour with what a healthy unit would do, and flag the drift long before a breakdown.

1.2 Problem Statement

A conventional diode-bridge rectifier with a capacitive filter draws a non-sinusoidal input current, resulting in reduced power factor and increased current harmonics. In addition, gradual degradation of the DC-link capacitor can increase output-voltage ripple and affect the performance and reliability of a power converter. The problem addressed in this project is to study the operation and performance of a single-phase boost Power Factor Correction (PFC) converter using MATLAB/Simulink and to investigate an artificial neural network (ANN)-based method for monitoring DC-link capacitor degradation. The proposed simulation-based method

estimates the capacitance from features extracted from the output-voltage waveform and identifies potential degradation by comparing the estimated capacitance with a predefined threshold. A low-voltage hardware prototype using the same basic converter topology is also retained to demonstrate the practical implementation of the power stage. The overall objective is to combine converter performance analysis with a computationally simple approach to capacitor condition monitoring.

1.3 Objectives

- To study the operation and performance of a single-phase boost Power Factor Correction (PFC) converter using MATLAB/Simulink and analyse its input and output electrical characteristics.
- To simulate the converter and examine important parameters such as DC-link voltage, inductor current, output-voltage ripple, power factor, and current total harmonic distortion (THD), wherever applicable to the model.
- To scale the design to a 10 W, 12 VAC to 24 V DC prototype using the UCC28180 CCM PFC controller.
- To develop an artificial neural network (ANN)-based capacitor condition-monitoring method using simulated output-voltage waveforms, estimate the DC-link capacitance from extracted waveform features, and detect potential degradation using a predefined capacitance threshold.
- To compare theoretical and simulated converter results where applicable, evaluate the ANN's capacitance-estimation performance, and analyse the losses that limit efficiency at low voltage.

1.4 Design Specifications

The specifications of the full-scale design and of the scaled prototype are listed in Table 1.1. The prototype uses a boost ratio close to that of the full-scale design, so the duty cycle varies over the line period in almost the same way and the control behaviour carries over directly.

Table 1.1: Design specifications of the full-scale design and the hardware prototype

Parameter	Simulation	10 W prototype (hardware)
Input voltage	230 V_{rms}, 50 Hz (325 V peak)	12 V_{rms}, 50 Hz (17 V peak)
Output voltage	400 V DC	24 V DC
Rated output power	303 W	≈ 10 W (9.6 W at 60 Ω)
Peak-input / output ratio	0.81	0.71
Switching frequency	50 kHz	≈ 65 kHz
Topology	Single-phase boost PFC, CCM	Single-phase boost PFC, CCM
Control	Average current mode (PI blocks, multiplier, PWM)	Average current mode (UCC28180)
Target power factor	≥ 0.98	≥ 0.95
Target current THD	< 5 %	< 10 %
Output ripple target	≤ 2 % (8 V)	< 1 V peak-to-peak
Load	48.5 Ω	60 Ω (2 × 120 Ω in parallel)
Monitoring	ANN-based capacitance estimation (simulation)	Sensing points for V, I and temperature

1.5 Organisation of the Report

Chapter 2 reviews the literature on PFC topologies, control methods and condition monitoring of power converters. Chapter 3 presents the system architecture, the operating principle, the design calculations for both power levels and the control strategy. Chapter 4 describes the Simulink model and the simulation results, including the behaviour under varying line and load and the capacitor-monitoring study. Chapter 5 covers the hardware prototype, its bill of materials, wiring and test procedure. Chapter 6 compares theoretical and simulated values and discusses the results. Chapter 7 concludes the report and outlines future work.

CHAPTER 2: LITERATURE SURVEY

2.1 Review of Related Work

Singh et al. [1] classified the large family of single-phase AC–DC converters with improved power quality and showed that the boost converter placed after a diode bridge is the most widely used front-end because its input inductor naturally smooths the line current. García et al. [4] surveyed single-phase PFC solutions and pointed out that CCM operation with average current control gives the lowest distortion at medium and high power, whereas discontinuous or critical conduction is preferred only below a few hundred watts.

For EV charging in particular, Yilmaz and Krein [2] reviewed charger topologies and power levels and identified the two-stage arrangement, a PFC rectifier followed by an isolated DC–DC converter, as the standard for Level 1 and Level 2 on-board chargers. Khaligh and Dusmez
[18] compared conductive and inductive charging architectures and highlighted the trade-off between component count and efficiency. To reduce bridge losses at kilowatt levels, Huber et al. [3] evaluated bridgeless PFC boost rectifiers, and Musavi et al. [17] proposed an interleaved bridgeless converter for plug-in hybrid chargers that improves efficiency at light load. These topologies are more efficient but need more switches, more complex sensing and careful handling of common-mode noise.

On the control side, the classic Unitrode application note by Todd [5] set out the design procedure for average-current-mode PFC around the UC3854, with an analogue multiplier fed by the rectified line voltage. Newer controllers such as the UCC28180 [6] remove the line-sensing network and generate the current reference internally from the output voltage and duty cycle, which reduces the external part count to a handful of resistors and capacitors. Erickson and Maksimovic´ [7] and Mohan et al. [8] provide the small-signal models used to tune the inner current loop and the slow outer voltage loop.

Reliability studies form the second strand of the survey. Yang et al. [10] reported from an industry survey that capacitors, semiconductor devices and solder joints are the leading causes of converter failure. Wang and Blaabjerg [9] reviewed DC-link capacitor reliability and noted that the electrolytic capacitor’s lifetime roughly halves for every 10 °C rise in core temperature, with end-of-life commonly declared when capacitance falls by 20 % or ESR doubles. Soliman et al.
[20] and Vogelsberger et al. [19] described methods that estimate capacitance or ESR online from voltage and current ripple. Tao et al. [11] defined the digital twin as a continuously updated virtual counterpart of a physical asset, and Zhao et al. [21] reviewed artificial-intelligence applications in power electronics, observing that most published work targets design optimisa-tion and control, while lifetime prediction from field data is still emerging.

For learning-based anomaly detection, LSTM encoder–decoder networks [13], [14] and physics-informed neural

networks [12] have been proposed as ways to learn normal behaviour from limited data. Table 2.1 summarises the most relevant works and what each contributes to this project.

Table 2.1: Summary of the reviewed literature

Ref.	Focus	Key finding used in this work	Gap relevant to this project
[1], [4]	Survey of single-phase PFC convert-ers	CCM boost with average cur-rent control gives the lowest THD at kW level	No condition monitoring
[2], [18]	EV charger topolo-gies and power levels	Two-stage OBC: PFC + iso-lated DC–DC; 3.3 kW is a common Level 2 rating	Reliability treated quali-tatively
[3], [17]	Bridgeless / inter-leaved PFC	Removes one bridge diode drop; better efficiency	More switches and sens-ing; harder to prototype
[5], [6]	UC3854 and UCC28180 con-trollers	Design procedure; UCC28180 needs no line sensing	Hardware focus only
[9], [10]	Converter and capac-itor reliability	Capacitors and devices fail most; 10 °C rule; 20 % C-loss end-of-life	No online prognosis
[19], [20]	Capacitor condition monitoring	C and ESR can be estimated from ripple	Not combined with learning-based health index
[11], [21]	Digital twins; AI in power electronics	Twin concept; AI mostly used for design and control	Few RUL studies on chargers
[12]–[14]	LSTM-AE and PINN models	Learn healthy behaviour; detect drift by reconstruction error	Not applied to OBC front-ends

2.2 Comparison of Topologies / Methods

Candidate PFC front-ends for a 3.3 kW charger are compared in Table 2.2. The figures are typical values reported in the cited literature rather than measurements from this project.

Table 2.2: Comparison of single-phase PFC front-end topologies for a 3.3 kW charger

Topology	Active switches	Semiconductors in conduction path	Typical efficiency	Control / sensing complexity	Prototype suitability
Passive (bridge + C + choke)	0	2 diodes	High, but PF ≈ 0.6–0.7	None	Fails harmonic limits
Conventional boost PFC (CCM)	1	2 bridge diodes + switch or diode	95–97 %	Low; dedicated ICs available	Best
Interleaved boost PFC	2	2 bridge diodes + switch or diode	96–97.5 %	Moderate; phase balancing	Moderate
Bridgeless boost PFC	2	1 diode + switch or diode	97–98 %	High; AC-side current sensing, EMI	Difficult
Totem-pole (GaN/SiC)	2–4	1–2 switches	98–99 %	High; digital control, fast devices	Difficult

Methods for monitoring converter health are compared in Table 2.3.

Table 2.3: Comparison of condition-monitoring approaches

Approach	Data needed	Strength	Limitation
Reactive protection (thermal trip, OVP, OCP)	Threshold signals	Simple, fast, certified	Acts only after a fault; no warning
Scheduled replacement	Operating hours	No sensing needed	Replaces healthy parts; misses early failures
Model-based parameter estimation	waveform features	Physically interpretable (C, ESR, R_{th})	Needs accurate model; sensitive to operating point
Data-driven anomaly detection (autoencoder, LSTM-AE)	Multichannel time series	Learns normal behaviour without fault labels	Opaque; needs healthy training data
ANN-based capacitance estimation (this work)	V, I, ripple, temperatures + physics model	Simple feedforward network; estimates capacitance from waveform features	Trained on simulated data; not yet validated on hardware measurements
Output-voltage			

2.3 Inference from the Survey

The survey leads to three design decisions. First, the conventional CCM boost PFC is the right front-end for this project: it meets the power-quality targets at 3.3 kW, it can be built safely at low power with a single controller IC, and its behaviour scales cleanly between the two power levels. Second, average-current-mode control is preferred over peak-current or hysteretic control because it gives low distortion at a fixed switching frequency, which simplifies filtering. Third, existing chargers achieve high conversion efficiency and protect themselves reactively, but they do not provide component-level predictive maintenance. Combining a physics model of the converter with a learning-based detector addresses that gap, because the physics model removes the influence of load and ambient temperature from the measurements and leaves the learning model to recognise genuine drift.

CHAPTER 3: PROPOSED SYSTEM AND DESIGN

3.1 Block Diagram

The proposed system combines a single-phase boost Power Factor Correction (PFC) converter with an artificial neural network (ANN)-based capacitor condition-monitoring method. The converter's electrical behaviour is studied using MATLAB/Simulink, while the DC-link capacitance is varied to generate output-voltage waveforms under different capacitor conditions. Relevant waveform features are extracted and supplied to a trained neural network to estimate the capacitance. The estimated value is compared with a predefined threshold to identify potential capacitor degradation. A low-voltage hardware prototype is retained to demonstrate the practical converter topology, while ESP32-based monitoring and cloud digital-twin functionality are considered possible future extensions.

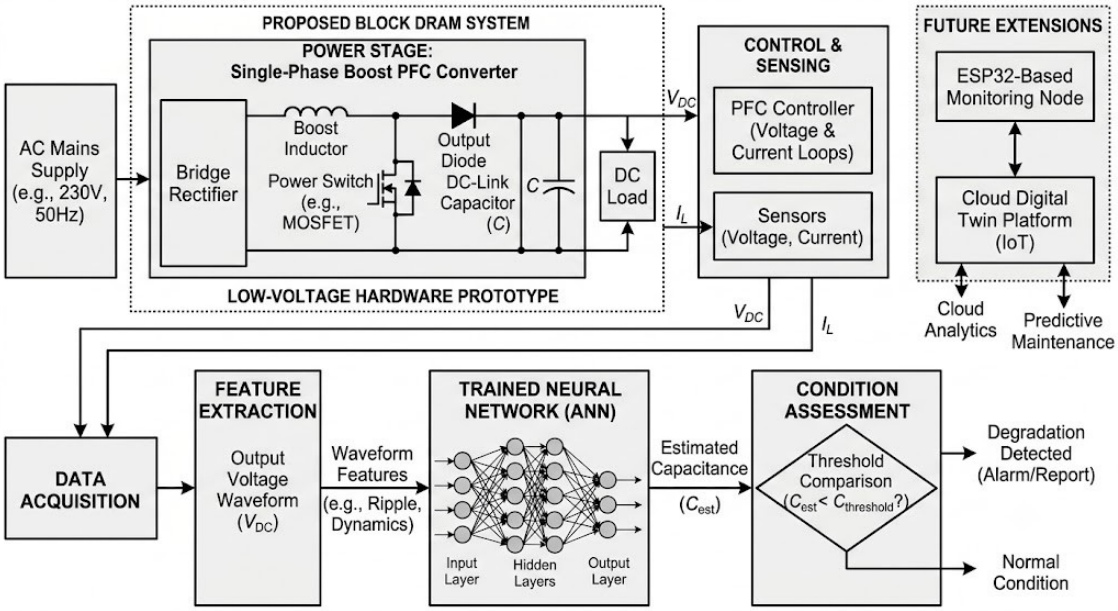


Figure 3.1: System block diagram of the AI-assisted charger front-end with the ANN-based monitoring layer

3.2 Circuit Topology

The power circuit is drawn in Figure 3.2. Diodes D_1–D_4 form the full-wave bridge. The boost inductor L, the MOSFET Q and the boost diode D make up the converter, and C_o with R_L represents the DC link and the load. A bypass diode D_{byp} connects the rectified input directly to the output; it conducts only while the output capacitor charges at switch-on, keeping the large inrush current away from the inductor and the boost diode, and remains reverse-biased once the output exceeds the input peak. The current-sense shunt R_{sh} is placed in the return path between the bridge and the ground rail, so the controller sees a voltage that is

proportional to the inductor current and negative with respect to its ground.

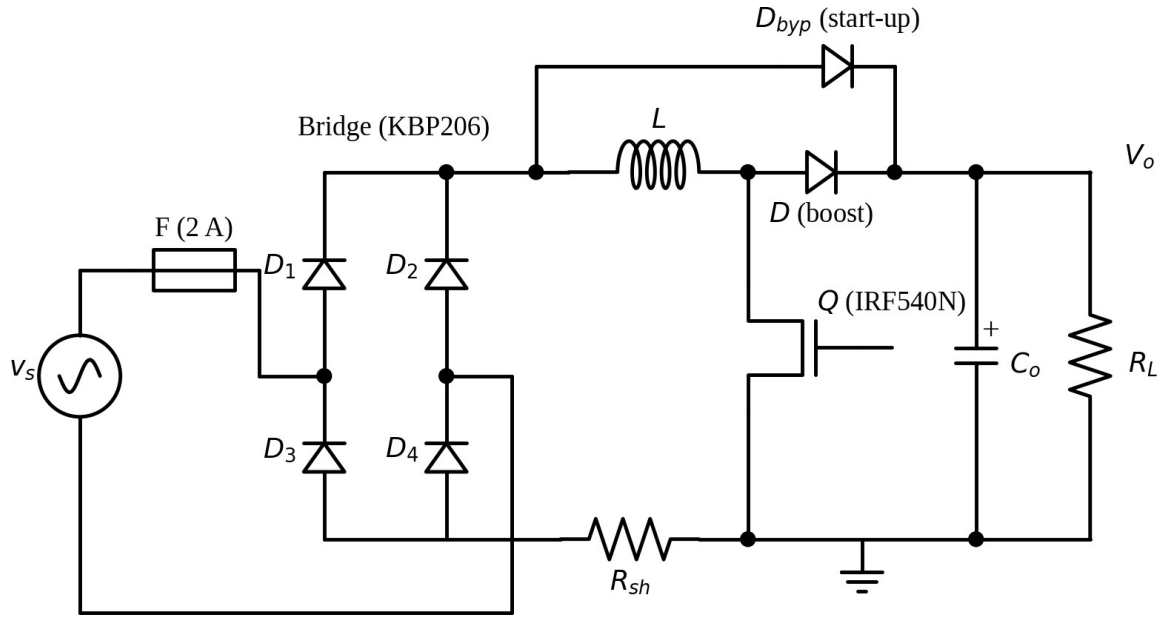
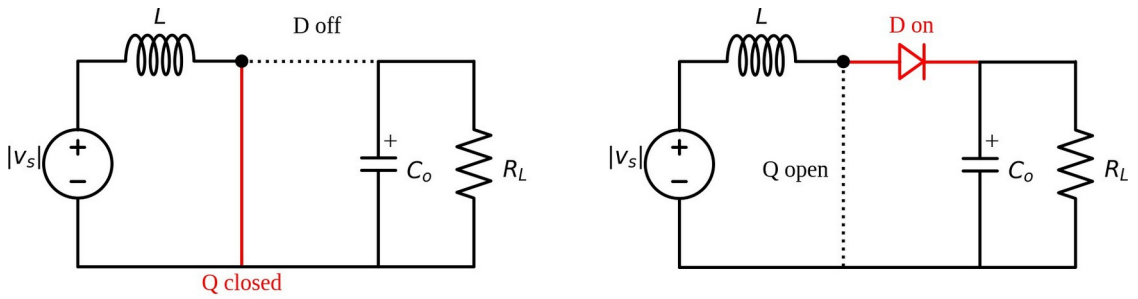


Figure 3.2: Single-phase boost PFC power circuit with bridge rectifier, bypass diode and current-sense shunt

3.3 Principle of Operation

The MOSFET switches at a frequency far above the 50 Hz line, so within one switching period the rectified input $|v_s|$ can be treated as constant. The two switching states are shown in Fig-ure 3.3. When Q is on, the boost diode is reverse-biased and the inductor is connected across the rectified input, so its current rises at the rate $|v_s|/L$ while the capacitor alone supplies the load. When Q is off, the inductor current continues through D into the capacitor and load; because V_o is larger than $|v_s|$, the current falls at the rate $(V_o - |v_s|)/L$.



(a) Switch ON: L charges

(b) Switch OFF: L discharges

Figure 3.3: Equivalent circuits of the boost converter during (a) the switch-on interval and (b) the switch-off interval

3.4 Control Strategy

Average-current-mode control uses two nested loops, as shown in Figure 3.4. The outer voltage loop compares V_o with its reference and passes the error through a PI controller with output limits; its output A is the amplitude of the current the converter must draw to balance input and output power. A multiplier shapes this amplitude with the normalised rectified input $|v_{in}|/V_{pk}$, producing the sinusoidal current reference i_{ref}. The inner current loop compares the measured inductor current with i_{ref}, and its PI output is the duty cycle d, which a PWM comparator converts into gate pulses.

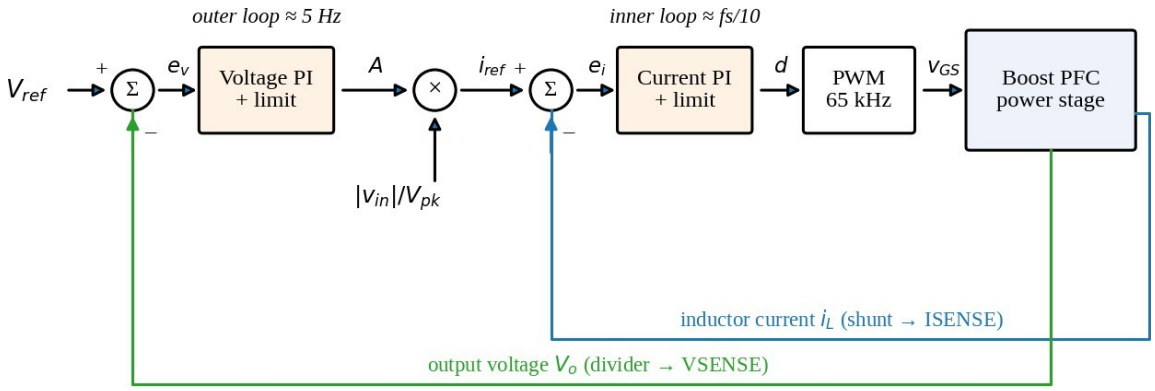


Figure 3.4: Average-current-mode control structure with outer voltage loop, multiplier and inner current loop

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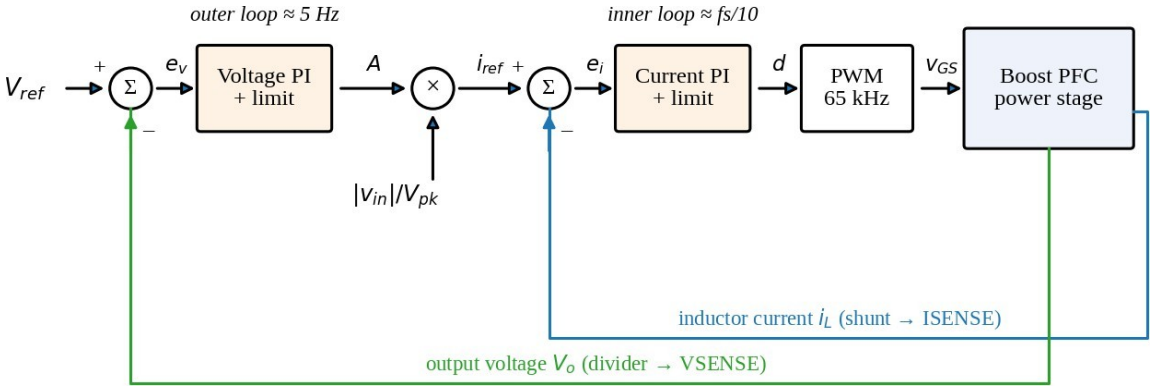


Figure 3.4: Average-current-mode control structure with outer voltage loop, multiplier and inner current loop

The two control loops are designed to operate at different bandwidths to ensure stable converter operation. The inner current-control loop responds rapidly to changes in the inductor current by adjusting the switching duty cycle. In the 300 W simulation model, the current-loop crossover frequency is selected as 5 kHz, which is one-tenth of the switching frequency. The proportional and integral gains are selected to provide a fast current response while maintaining stable operation.

The outer voltage-control loop regulates the DC-link voltage by adjusting the reference amplitude of the input current. Since the DC-link capacitor stores energy, changes in the charging current affect the output voltage gradually. The voltage loop is therefore designed with a much lower bandwidth than the current loop. Its crossover frequency is approximately 5 Hz, which is significantly below the 100 Hz twice-line-frequency ripple. This separation helps prevent the voltage controller from responding excessively to the natural DC-link voltage ripple and reduces unwanted distortion in the input current.

In the hardware prototype, the corresponding control functions are implemented internally by the UCC28180 controller. The transconductance voltage amplifier and VCOMP compensation network regulate the output voltage, while the internal current-control circuitry and ICOMP compensation network regulate the inductor current. The controller generates the input-current reference internally from the duty cycle and output voltage, eliminating the need for a separate rectified line-voltage sensing divider in the prototype.

CHAPTER 4: SIMULATION STUDIES

4.1 Simulation Model

The 300 W converter was built in MATLAB/Simulink with Simscape Electrical components; the model is reproduced in Figure 4.1. The power path consists of an AC voltage source, a four-diode bridge, the boost inductor, the MOSFET with its body diode, the boost diode, and the output capacitor with a resistive load. A second diode from the rectified bus to the output acts as the start-up bypass. Voltage and current measurement blocks provide the input voltage and current, the rectified voltage, the inductor current and the output voltage. The control section mirrors Figure 3.4: a constant of 400 feeds a summing junction, a PI block and a saturation block that form the outer voltage loop; a gain and multiplier generate the sinusoidal current reference from the rectified voltage; a second summing junction, PI block and saturation block form the inner current loop; and a PWM generator with a gain of 10 drives the gate. The f(x)
= 0 block is the Simscape solver configuration, and the scopes display the input voltage and current together for the power-factor check. The simulation stop time is 0.4 s.

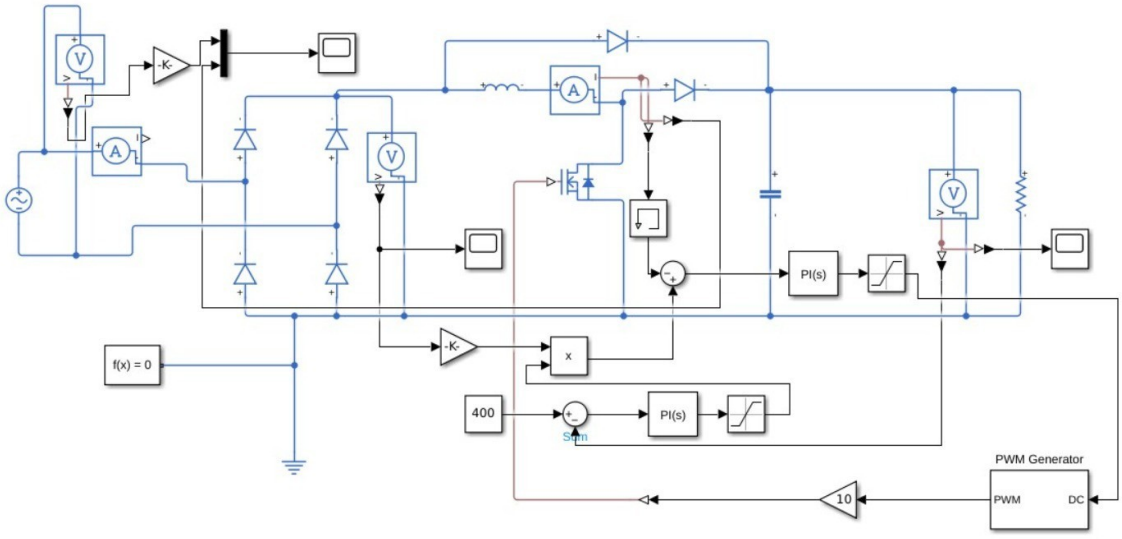


Figure 4.1: MATLAB/Simulink model of the 300 W boost PFC converter with average-current-mode control

To obtain switching-level waveforms with exactly known parameters and to run many operating points quickly, the same circuit and control structure were also coded as a fixed-step switching model in Python (Appendix A). It includes the bridge and diode forward drops, the switch on-resistance and the inductor resistance, and it uses the gains calculated in Section 3.5. The parameters common to both models are listed in Table 4.1. The 10 W prototype was simulated in the same way with the actual part values, so that the oscilloscope waveforms can be compared against a reference.

4.2 Simulation Results

Figure 4.2 compares the input voltage and input current of the boost PFC converter under two operating conditions: with power factor correction disabled and with the PFC controller enabled. When the controller is disabled, the bridge rectifier and output capacitor draw a non-sinusoidal, pulsed input current. When the controller is enabled, the boost PFC stage shapes the input current to follow the input voltage more closely, improving the power factor and reducing current harmonics. The comparison demonstrates the role of PFC control in improving input-current quality. In simulation, the 300 W design achieves a power factor of 0.997 and a current THD of 4.0 %.

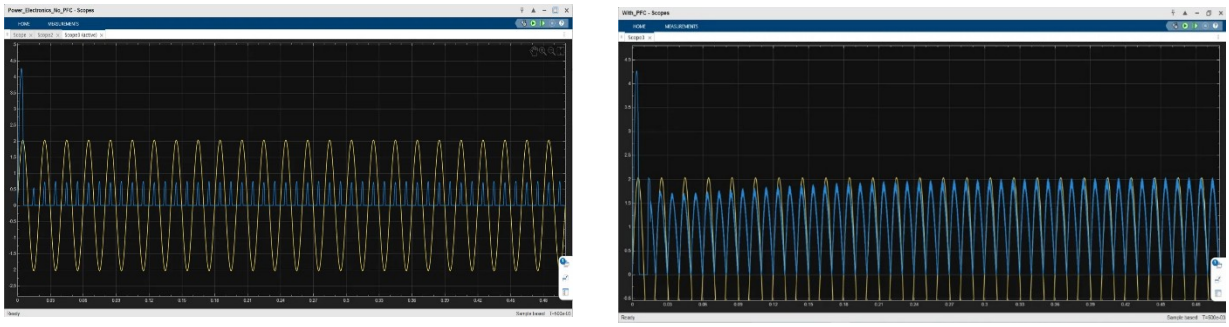


Figure 4.2: Line voltage and current of the simulation stage: (a) diode rectifier without PFC, (b) with average-current-mode PFC

The start-up of the output voltage is shown in Figure 4.3. The bypass diode pre-charges the capacitor to about 323 V, close to the input peak. The voltage loop then ramps the output to the 400 V reference, which it reaches within 2 % in 93 ms with an overshoot to only 405 V (1.3 %). In steady state the 100 Hz ripple is 8.1 V peak-to-peak, or 2.0 % of the output, which agrees with the 8 V used to size C_o in (3.11).

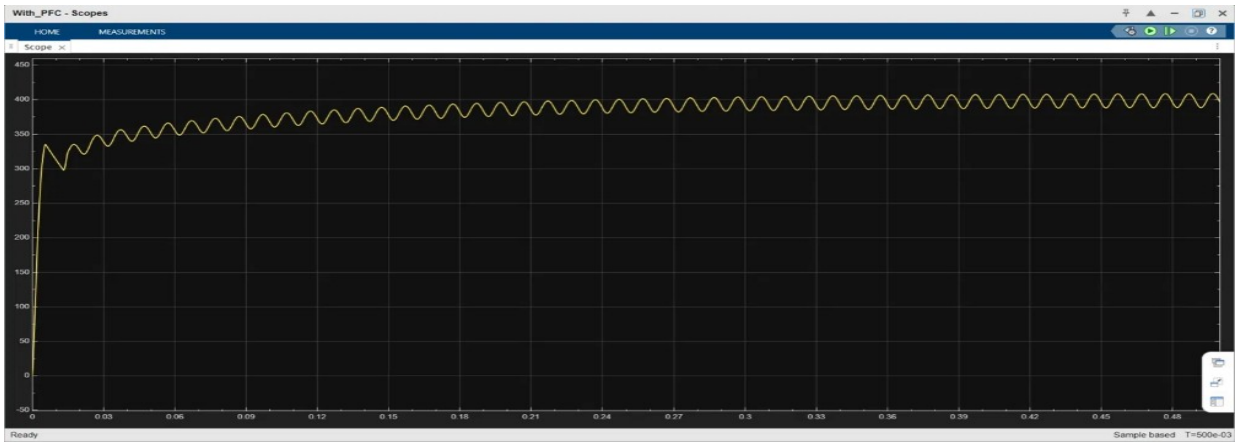


Figure 4.3: Output voltage of the simulation stage

4.2.1 Scaled 10 W prototype

The 10 W prototype was simulated with its real components (12 VAC, 330 µH, 2200 µF, 60 Ω, 65 kHz). Figure 4.6 shows the input voltage and current with the controller off and on. Without PFC the current again flows in narrow spikes and the power factor is 0.58. With the UCC28180 running, the current follows the voltage as a sinusoid and the power factor is 0.99. Small flat segments appear in the current at each zero crossing: below about 1.7 V the bridge diodes cannot conduct, so the current stays at zero for a short time. This dead band is a low-voltage effect; at 230 V it occupies a negligible fraction of the cycle.

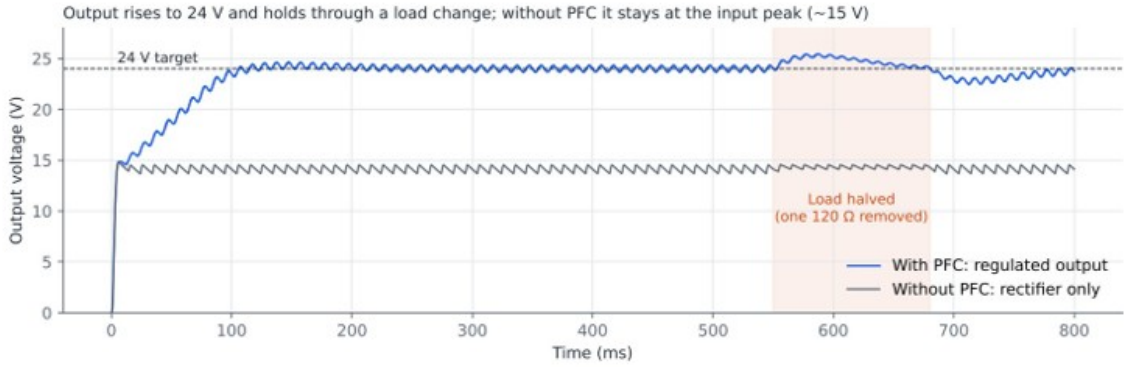


Figure 4.4: Output voltage of the 10 W prototype: start-up, regulation and a load change, with and without PFC

Figure 4.8 shows the inductor current near the AC peak at the switching scale. The current ripples by about 0.3 A peak-to-peak at 65 kHz around the sinusoidal reference, in line with the 0.23–0.28 A calculated in Section 3.4.2.

CHAPTER 5: HARDWARE IMPLEMENTATION

5.1 Scaling Rationale

Testing a 400 V, 300 W converter on an open bench is unsafe for a student laboratory, so the hardware demonstrates the same topology and control at low voltage and power. The concept stage had considered a 100 W, 24 VAC to 48 V prototype. That level would have needed an input current of about 4 A RMS, heavier wiring and a soldered layout, and its 48 V bus would sit just under the extra-low-voltage limit with no margin for overshoot. The final prototype was therefore scaled to 10 W, 12 VAC to 24 V DC, which allows solderless construction on a breadboard. Its peak-input to output ratio of 0.71 is close to the 0.81 of the full design, so the duty cycle sweeps a similar range over each line half-cycle. Scaling changes component values but not behaviour: according to (3.8) and (3.11), a lower voltage needs a smaller inductance and a larger capacitance for the same relative ripple.(3.8)(3.11),(3.8)(3.11),

5.2 Circuit Description

The hardware circuit is shown in Figure 5.1. Solid lines carry the power current of about 1 A and are wired with doubled jumpers; dashed lines are control signals. The second wire of the AC supply returns to the other AC terminal of the bridge. The MOSFET source, the capacitor negative, the load and pin 1 of the UCC28180 meet on the ground rail on the load side of the shunt, so that the shunt voltage seen by ISENSE contains only the inductor current. A PFC on/off switch in the VCC line lets the controller be disabled to demonstrate the uncorrected waveform.

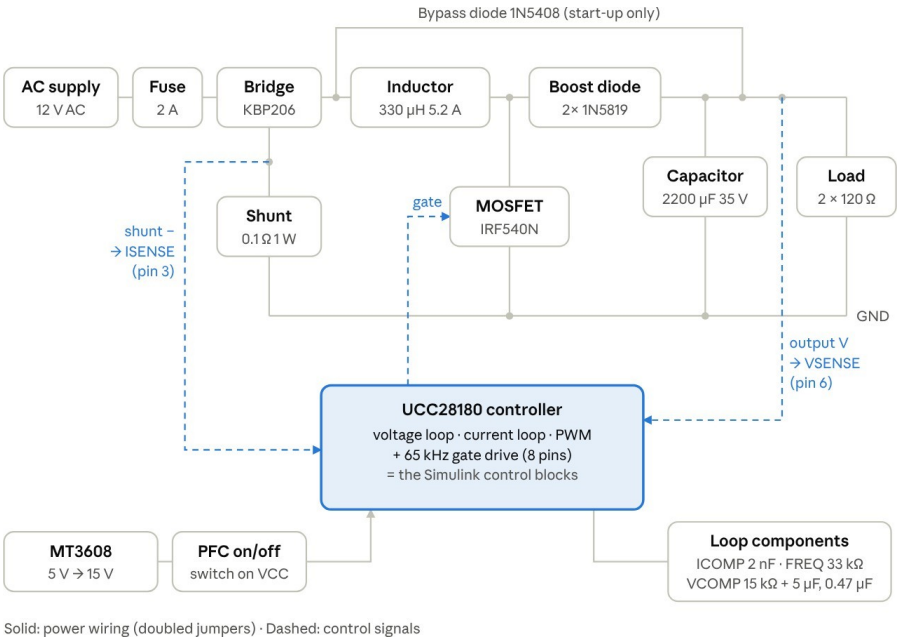


Figure 5.1: Circuit of the 10 W prototype: power stage and UCC28180 control connections

Each hardware block maps onto a block of the Simulink model, as listed in Table 5.1. The only block without a hardware counterpart is the rectified-voltage sensor that feeds the Simulink multiplier, because the UCC28180 derives its current reference internally.

Table 5.1: Correspondence between hardware blocks and Simulink blocks

Hardware block	Simulink equivalent	Function
Lab AC supply + 2 A fuse	AC voltage source	12 VAC, 50 Hz input; fuse clears a short circuit
KBP206 bridge	Four diodes	Full-wave rectification to 0–17 V half-sines
330 µH toroid	Inductor	Stores energy while Q is on, re-leases it at higher voltage
IRF540N MOSFET	MOSFET	Boost switch; duty cycle sets boost and current shape
2 × 1N5819	Boost diode	Unidirectional path from inductor to output
1N5408	Top (bypass) diode	Pre-charges the capacitor at switch-on
2200 µF capacitor	Capacitor	Smooths the output to steady 24 V
2 × 120 Ω	Resistor	Load of about 9.6 W
0.1 Ω shunt	Current measurement	Converts inductor current to a small negative voltage
Divider into VSENSE	Output voltage measure-ment	Scales 24 V to the 5 V reference
Voltage amplifier + VCOMP network	Constant 400, PI(s), satu-ration	Sets the current demand from the output error
Current loop + ICOMP capacitor	Multiplier, PI(s), satura-tion	Makes the average inductor current sinusoidal
Oscillator + gate driver	PWM generator and gain	Produces the ≈ 65 kHz gate pulses
MT3608 module	—	Boosts 5 V USB to 15 V for the controller

5.3 Bill of Materials

The components actually used in the prototype are listed in Table 5.2 and Table 5.3. The UCC28180 is supplied only in an SOIC-8 package, so it is soldered once onto an SOIC-to-DIP adapter with header pins; every other part is inserted on an 830-point breadboard.

Table 5.2: Power-stage components of the prototype

Component	Part and rating	Qty
Input supply	Laboratory AC power supply, 12 VAC, 50 Hz	1
Fuse and holder	5 × 20 mm 2 A glass fuse in BF-013A holder	1
Bridge rectifier	KBP206 (600 V, 2 A)	1
Boost inductor	330 µH, 5.2 A toroidal	1
MOSFET	IRF540NPBF (100 V, 33 A, 44 mΩ) on TO-220 heatsink	1
Boost diode	1N5819 Schottky (40 V, 1 A), two in parallel	2
Bypass diode	1N5408 (1 kV, 3 A)	1
Output capacitor	2200 µF, 35 V electrolytic, 105 °C	1
Current shunt	0.1 Ω, 1 W	1
Load	120 Ω, 10 W wire-wound, two in parallel (60 Ω)	2

Table 5.3: Control-circuit components of the prototype

Component	Part and value	Qty
PFC controller	TI UCC28180DR on SOIC-8 to DIP-8 adapter with header pins	1
Controller supply	MT3608 boost module set to 15 V, fed from a USB 5 V source	1
Resistors	220 Ω, 33 kΩ × 2, 4.7 kΩ, 10 kΩ × 2, 15 kΩ, 10 Ω	9
Capacitors	1 nF polyester × 4, 0.47 µF polyester, 10 µF × 2 (in series), 1 µF, 0.22 µF	9
Clamp and gate diodes	1N5408 (ISENSE clamp), 1N5819 (gate turn-off)	2
PFC on/off switch	SPST toggle switch in the VCC line	1
Construction	MB102 830-point breadboard, jumpers, crocodile-clip leads	1 set

The components were purchased from Robu.in; the order is itemised in Table 5.4. Some lines include spares, and a few small capacitors (150 pF, 10 pF and the tantalum parts) were ordered for an earlier controller option and are not used in the final circuit. Resistors, the toggle switch and the breadboard were bought separately, and the AC supply, oscilloscope and multimeter were provided by the laboratory.

Table 5.4: Itemised component order (Robu.in)

Item	Qty	Unit price (Rs.)	Total (Rs.)
KBP206 bridge rectifier, 600 V 2 A	2	13.00	26.00
IRF540NPBF N-MOSFET, 100 V 33 A, TO-220	2	58.00	116.00
1N5819 Schottky diode, 40 V 1 A	6	10.00	60.00
1N5408 rectifier diode, 1 kV 3 A	2	21.00	42.00
150 pF 50 V ceramic capacitor (spare)	24	0.45	10.80
BF-013A fuse holder	3	14.00	42.00
5 × 20 mm 250 V 2 A glass fuse	3	9.00	27.00
2200 µF 35 V electrolytic capacitor	2	17.00	34.00
0.47 µF 310 VAC polyester box capacitor	2	11.00	22.00
1 µF 160 V electrolytic capacitor	2	8.00	16.00
MT3608 2 A DC–DC boost module	1	39.00	39.00
USB Type-C breakout board	1	99.00	99.00
Type-C 5 V 2 A step-up module	1	64.00	64.00
Crocodile clip set (6 pcs)	1	49.00	49.00
330 µH 5.2 A toroidal inductor	1	120.00	120.00
10 pF 50 V ceramic capacitor (spare)	21	0.48	10.08
0.22 µF ceramic capacitor	5	2.10	10.50
10 µF 50 V aluminium electrolytic (Rubycon)	8	8.00	64.00
1 nF 100 V polyester film capacitor	12	2.87	34.44
0.47 µF 35 V tantalum capacitor (spare)	2	15.00	30.00
SOIC-8 to DIP-8 adapter board (pack of 5)	1	39.00	39.00
UCC28180DR CCM PFC controller	2	89.00	178.00
2.54 mm 1 × 40 male header strip	2	9.00	18.00
TO-220 aluminium heatsink (pack of 5)	1	29.00	29.00
4.7 µF 16 V tantalum capacitor (spare)	2	13.00	26.00
Total (excluding resistors, shipping and taxes)			**1,205.82**

5.4 Controller Wiring

The connections around the UCC28180 are given in Table 5.5. The values follow the datasheet design procedure [6] scaled to 12 V → 24 V at 10 W, and their derivation is shown in Section 3.4.3. The shunt sits between the bridge negative terminal and the ground rail, and the MOSFET source, capacitor negative, load and pin 1 return to the ground rail.

Table 5.5: UCC28180 pin-by-pin wiring

Pin	Name	Connection
1	GND	Ground rail (load side of the shunt)
2	ICOMP	2 nF (two 1 nF polyester in parallel) to GND
3	ISENSE	220 Ω to shunt negative (bridge side); 1 nF to GND; 1N5408 clamp with anode at GND and cathode at pin 3
4	FREQ	33 kΩ to GND (≈ 65 kHz)
5	VCOMP	15 kΩ in series with 5 µF (two 10 µF in series) to GND; 0.47 µF to GND
6	VSENSE	33 kΩ + 4.7 kΩ in series from +24 V; 10 kΩ to GND; 1 nF to GND (set point 23.85 V)
7	VCC	MT3608 output (15 V); 1 µF and 0.22 µF to GND
8	GATE	10 Ω to MOSFET gate with 1N5819 across it (anode at gate); 10 kΩ gate to source

5.5 Construction Practice

Breadboard contacts add resistance and stray inductance, which matter in the switching loop formed by the MOSFET, the boost diode and the output capacitor. These three parts are there-fore placed in neighbouring rows so that the high-frequency current loop is as small as possible, and every power connection uses two jumpers in parallel. The 10 Ω gate resistor limits the turn-on current from the controller's 1.5 A driver and damps ringing in the gate loop, the anti-parallel 1N5819 speeds up turn-off, and the 10 kΩ gate-to-source resistor keeps the MOSFET off while the controller is unpowered. The controller decoupling capacitors are placed directly at pins 7 and 1. The MOSFET is mounted on a TO-220 heatsink, although its expected loss is only about 60 mW.

5.6 Test Procedure

The prototype is built and tested in the order given in Table 5.6, so that each fault appears at the step that introduced it.

Table 5.6: Build and test sequence

Step	Action	Pass criterion
1	Adjust the MT3608 to 15 V before con-necting it to the controller	15 V at UCC28180 pin 7
2	Power the controller alone, without AC input	Oscillator active at pin 4; gate pulses ≈ at 65 kHz on pin 8 (may be narrow or absent with no output voltage)
3	Wire the power stage with doubled jumpers and compact switching loop	Continuity and polarity checked with multimeter
4	With the controller unpowered, raise the AC supply slowly from 0 to 12 V	Output ≈ 14 V; input current in narrow pulses (record as “without PFC”)
5	Power the controller, then raise the AC supply slowly to 12 V	Output settles at ≈ 24 V; input current sinusoidal and in phase. If the output exceeds 28 V, reduce the supply and check the VSENSE divider
6	Remove one 120 Ω resistor and reconnect it	Output remains near 24 V
7	Run the Simulink model at 17 V peak, 24 V reference, 330 µH, 2200 µF, 60 Ω	Simulated and measured waveforms placed side by side

5.7 Measurement Setup and Safety

The oscilloscope connections are listed in Table 5.7. The AC-side method is used only if the laboratory supply output is isolated from earth, and only one ground-clip position is used at a time, because the two probe grounds of a bench oscilloscope are tied together and to earth. Power factor is computed from the AC-side voltage and current traces, and the ripple is read with AC coupling.

Table 5.7: Oscilloscope probing plan

Measurement	Probe tips	Ground clip	Time-base
Rectified voltage and inductor current	CH1: bridge (+); CH2: shunt negative (inverted; 0.1 V per A)	Ground rail	5 ms/div
AC voltage and AC current (PF figure)	0.1 Ω in one AC lead; CH1: other AC lead; CH2: far end of the 0.1 Ω (inverted)	AC lead–shunt junc-tion	5 ms/div

Gate pulses	MOSFET gate	Ground rail	10 µs/div
Output ripple	+24 V with AC coupling	Ground rail	5 ms/div

The prototype is demonstrated in five stages: the Simulink model with its overlapping input voltage and current, a walk-through of the board matching each part to its Simulink block,

the uncorrected waveform with the controller switched off, the corrected waveform with the controller on, and a load change showing that the output holds at 24 V.

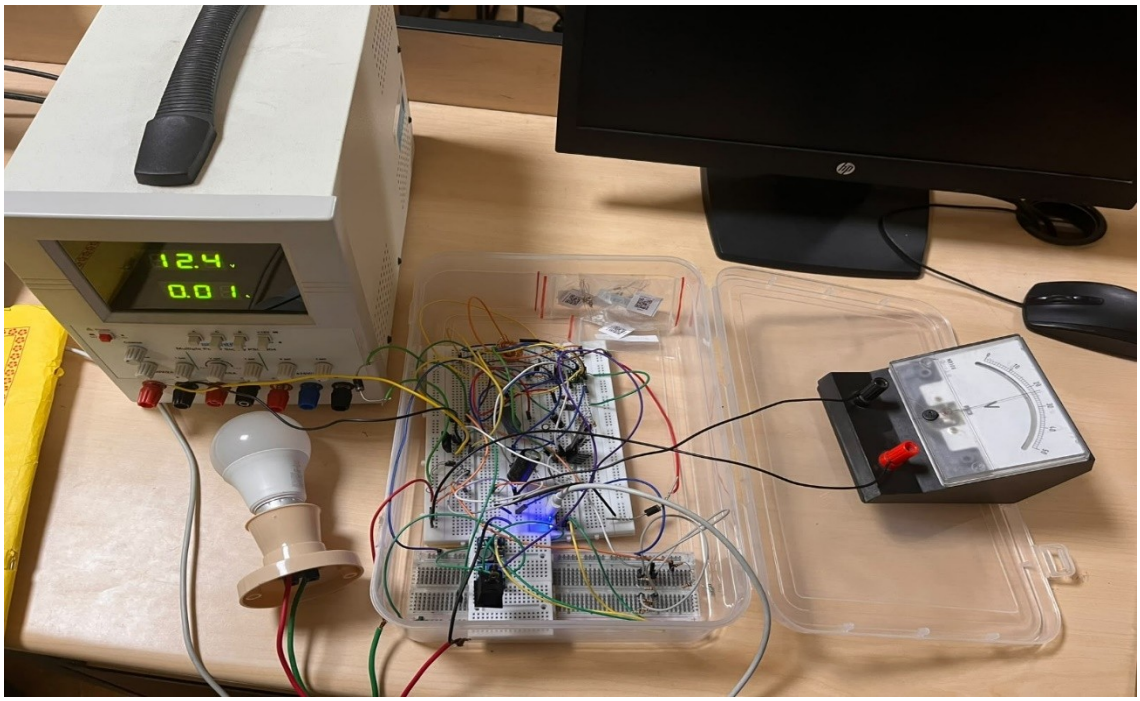


Figure 5.2: Circuit of the 10 W prototype: power stage and UCC28180 control connections

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Comparison of Calculated and Simulated Results

The calculated and simulated values agree closely: the output set point differs by less than 1 %, the 100 Hz ripple by 0.03 V, and the peak inductor current by 0.02 A. Bench measurements from the 10 W prototype are being consolidated and are not included in this version.

6.2 Discussion

Power quality. The simulated converter meets its power-factor target with margin: the 300 W design reaches a power factor of 0.997 with 4.0 % THD. The simulated prototype reaches 0.99, but its THD of 8.7 % is about twice that of the full design. The extra distortion comes from the bridge dead band near the zero crossings, where the 1.7 V forward drop is a tenth of the 17 V peak; at 230 V the same drop is less than 1 % of the peak and its effect vanishes.

Regulation and dynamics. The voltage loop was deliberately made slow, with a crossover near 5 Hz, so that the 100 Hz ripple does not modulate the current reference. The cost is a load-step excursion of about 6 % that settles in roughly 100 ms. For a two-stage charger this

is acceptable because the second-stage converter regulates the battery current independently of small DC-link variations.

Scaling. The calculated, simulated and expected hardware behaviour agree at both power lev-els, which indicates that the 10 W prototype reproduces the control behaviour of the
300 W design. What does not scale is efficiency and thermal behaviour; these were therefore evaluated analytically rather than on the prototype.

Predictive maintenance. The monitoring method was evaluated in simulation with synthetic capacitor ageing. The DC-link capacitance was varied from 120 µF to 60 µF across 41 simulation cases; five features were extracted from the output-voltage waveform (ripple, mean, standard deviation, maximum and minimum), and a feedforward neural network with eight hidden neurons was trained to estimate the capacitance. The estimated capacitance is compared with a 96 µF threshold, 80 % of the nominal value, to flag potential degradation. The study is a simulation with synthetic ageing, so the absolute values depend on the assumed degradation law; its value lies in showing that the pipeline works end to end and in fixing the sensing requirement of a ripple measurement accurate to about 2 %.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

A single-phase boost PFC front-end for a 300 W EV charger was designed, simulated and scaled to a 10 W hardware prototype, and an ANN-based capacitor condition-monitoring method was developed on top of it. With average-current-mode control the simulated 300 W stage converts 230 VAC to a regulated 400 V DC link with a power factor of 0.997, 4.0 % current THD and 2 % output ripple. The 10 W prototype uses the same topology with a UCC28180 controller; its design calculations and simulations predict a power factor of 0.99, 8.7 % THD, 0.6 V ripple and 82 % efficiency, with the efficiency limited mainly by bridge-diode losses that are specific to the low test voltage. The ANN-based method estimated the DC-link capacitance from output-voltage waveform features in simulation and flagged degradation against an 80 %-of-nominal threshold. Together these results show that power-quality performance and capacitor condition monitoring can be built into the same low-cost charger front-end.

7.2 Future Scope

- Add the isolated DC–DC stage (for example an LLC or dual-active-bridge converter) with CC/CV control to complete the charger.
- Fit the prototype with INA219 voltage/current sensors, NTC thermistors on the capacitor and heatsink, and an ESP32 publishing over MQTT/TLS, so that the condition-monitoring method runs on measured rather than synthetic data.
- Train and compare the LSTM autoencoder and a physics-informed neural network on accelerated-ageing tests of real capacitors in a thermal chamber.
- Move the 300 W design to a bridgeless or GaN totem-pole topology to raise efficiency above 98 %, and lay out a PCB for testing at higher power.
- Perform formal harmonic and EMC measurements against IEC 61000-3-2 and design the input EMI filter.

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APPENDICES

Appendix B: Component Datasheet Extracts

The ratings of the main parts that were used in the design checks are summarised in Table B.1.

Table B.1: Key ratings of the main components

Component	Key ratings	Value in circuit
UCC28180 [6]	18–250 kHz programmable; 5 V VSENSE reference; SOC −0.285 V, PCL −0.4 V at ISENSE; OVP at 107 % / 109 %; gate drive 1.5 A source / 2 A sink, clamped at 15.2 V	63.6 kHz; 23.85 V set point; 2.85 A / 4.0 A limits
IRF540NPbF [22]	V_{DS} 100 V; I_D 33 A; $R_{DS(on)}$ 44 mΩ at V_{GS} = 10 V; TO-220AB	24 V; 0.62 A RMS
KBP206	600 V repetitive reverse voltage; 2 A average; 1.1 V forward drop at 2 A	17 V peak; 0.98 A RMS
1N5819	40 V; 1 A average; Schottky	24 V; 0.2 A each (two in parallel)
1N5408	1000 V; 3 A average; 200 A surge	Start-up bypass; ISENSE clamp
330 µH toroid	5.2 A rated current	1.51 A peak
2200 µF electrolytic	35 V; ±20 %; 2000 h at 105 °C	24 V; 0.65 A RMS ripple
MT3608 module	Boost converter, up to 2 A switch current	5 V → 15 V for VCC